

Suitability of Transformer Core Materials

Investigation of Magnetic Hysteresis Loops

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Abstract

This report investigated the magnetic response of several transformer-core materials, including a CuNi alloy measured at two temperatures, by recording their B – H curves under the same alternating magnetic field. A 400-turn primary coil was driven with a 5 V_{p-p} 50 Hz current, this magnetised a 500-turn secondary coil. the induced current was passed through an integrator circuit before being read by a PicoScope converting the induced and driven voltages into H and B . Twenty steady-state loops were acquired for each sample, and the loop area together with the local dB/dH gradients yielded estimates of the relative permeability and the power loss per unit volume.

The air control data are approximately linear, but the measured $\mu_r = 0.37 \pm 0.03$ flags a large systematic calibration offset, so the absolute magnitudes should be treated with caution.

Among the ferromagnets, mild steel reached $\mu_{r,\max} \approx 1.2 \times 10^2$ with a hysteresis loss of about $1.38 \times 10^6 \text{ W m}^{-3}$, while transformer iron achieved $\mu_{r,\max} \approx 3.3 \times 10^2$ and only $4.73 \times 10^5 \text{ W m}^{-3}$ of loss. At 10 °C the CuNi alloy still exhibits a small loop ($\mu_{r,\max} \approx 7.9$, loss $\approx 4.6 \times 10^3 \text{ W m}^{-3}$), but heating it above 40 °C collapses the loop into a line with $\mu_r = 1.33 \pm 0.01$ and almost no hysteresis, consistent with it crossing a Curie-like transition.

Within the shared calibration uncertainties, transformer iron therefore emerges as the preferred material, mild steel sacrifices efficiency through its broader loop, and CuNi becomes weakly paramagnetic near 40 °C, rendering it unsuitable for transformer-core applications.

1 Introduction

Transformers form a core part of modern technology, being used for impedance matching within electronics and electricity transmission on the grid. In both systems, the efficiency and performance are determined by how the transformer responds to an applied magnetic field.

This response is described by the relationship between the magnetic flux density (B) and the magnetising field H , known as the B – H curve. By altering the transformer’s core, its response can be tuned. For ferromagnetic materials, the B – H curve is non-linear and exhibits hysteresis (**Figure 1**) due to magnetic subdomains in the material.

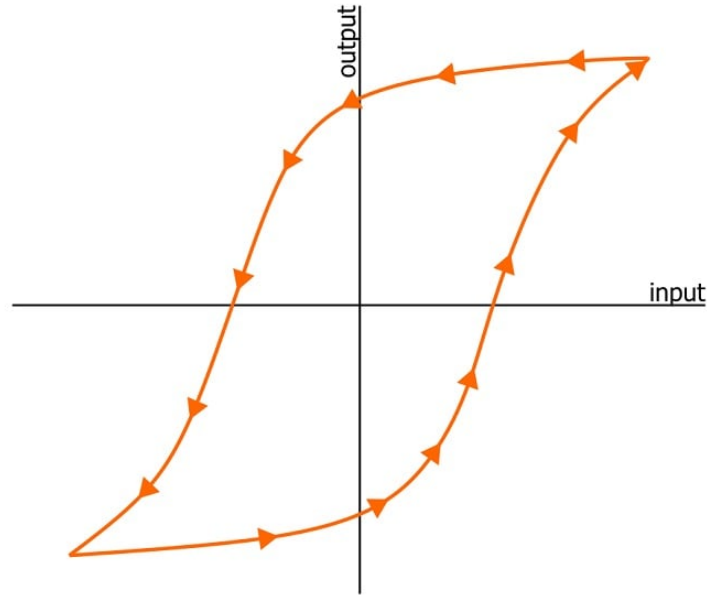


Figure 1: A typical hysteresis curve for a ferromagnetic material, showing the relationship between B and H . [1].

The most efficient transformers are made by having a high relative permeability (μ_r) and low energy loss per cycle. This experiment aims to quantify the usability of different materials as transformer cores by measuring the energy loss per cycle per unit volume (given by the area of a hysteresis loop) and the maximum magnetic susceptibility (given by the gradient). Section 2 of this report will go through the relevant theory. Section 3 outlines the experimental setup, and Section 4 shows the results obtained. This data is analysed and compared against theoretical expectations in Section 5. The overall conclusions are then presented in Section 6.

2 Theoretical Background

2.1 What Is Hysteresis

Hysteresis is the phenomenon where the value of a physical property lags behind changes in the effect causing it. In magnetic materials, this is observed as a lag between the magnetisation of the material and the external magnetic field applied to it. This results in a hysteresis loop when plotting magnetic flux density (B) against magnetic field strength (H). There are three key points on the hysteresis loop; these are shown in Figure 2.

- **(A) Saturation magnetisation (B_{sat}):** This is the maximum magnetic flux density that the material can achieve. Beyond this point, increasing the external magnetic field does not significantly increase B as the magnetic domains are fully aligned and thus $\mu_r \rightarrow 1$.
- **(B) Remanent magnetisation (B_r):** This is the magnetic flux density that remains in the material when the external magnetic field is reduced to zero.
- **(C) Coercive field (H_c):** This is the magnitude of the reverse magnetic field that must be applied to reduce the magnetic flux density to zero after the material has been magnetised to saturation.

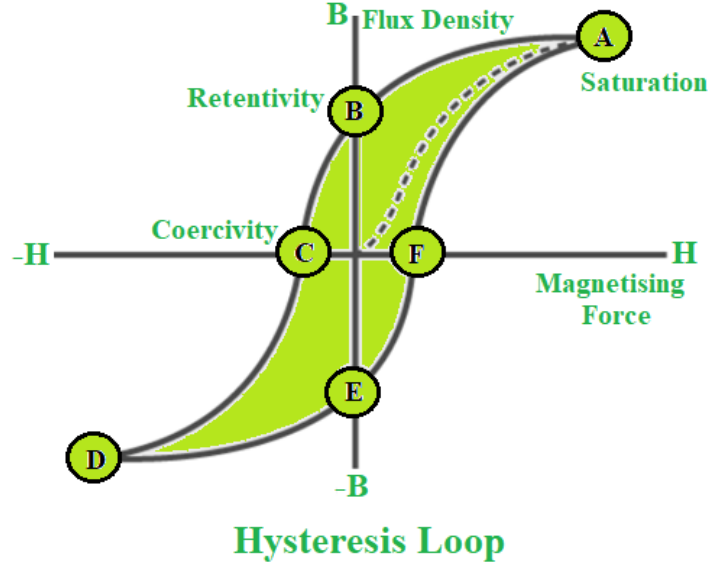


Figure 2: Labelled hysteresis loop [2].

Hysteresis is caused by the magnetic domains within the material. When an external magnetic field is applied, these domains tend to align with the field, increasing the overall magnetisation of the material. However, due to imperfections in the material and interactions between domains, they do not realign instantaneously. This time delay is visible because when the external field is changed, the magnetic domains are misaligned and have to absorb energy to realign themselves, causing energy loss in the form of heat.

2.2 Transformer Operation

The magnetising field H inside a solenoid is given by

$$H = \frac{nI}{L} \quad (2.1)$$

Here, n is the number of turns, I the current, and L the solenoid length.

This gives rise to the magnetic flux density B . For solenoids with non-magnetic media in their core, there is a linear relationship:

$$B = \mu_0(H + M) \quad (2.2)$$

$$M = \chi H \Rightarrow B = \mu_0(1 + \chi)H \quad (2.3)$$

$$\mu_r = 1 + \chi \Rightarrow B = \mu_0\mu_r H \quad (2.4)$$

However, when using magnetic materials, μ_r is not a constant and so:

$$B = \mu_0(H + M) \quad (2.5)$$

μ_0 is the permeability of free space ($4\pi \times 10^{-7} \text{ H/m} \approx 1.257 \times 10^{-6} \text{ H/m}$).

μ_r is the relative permeability of the material.

M is the magnetisation of the core material.

χ is the volume magnetic susceptibility.

By using different materials within the centre of the solenoid, different hysteresis loops formed as a result of the different values of μ_r . These B and H values cannot be measured directly, but if you arrange two solenoids as in **Figure 3**:

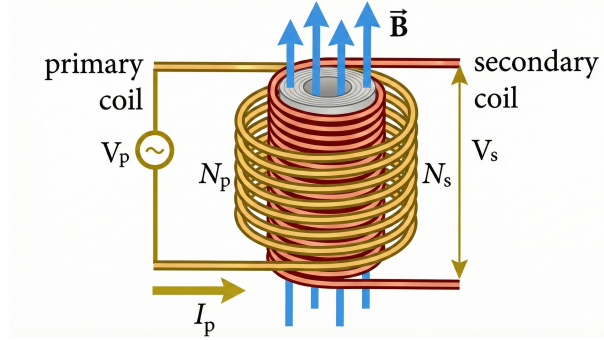


Figure 3: Interlinked solenoids.

The variance of the magnetic field in the primary solenoid results in an induced *emf* in the secondary coil. This is given by Faraday's law of induction:

$$\mathcal{E} = -N_s \frac{d\Phi}{dt} \quad (2.6)$$

Where:

$\mathcal{E}$ is the *emf* induced in the secondary coil

N_s is the number of turns of coil

$\frac{d\Phi}{dt}$ is the rate of change of flux

Φ is the magnetic flux, is given by:

$$\Phi = BA_s \quad (2.7)$$

Where B is the magnetic flux density inside the coil and is provided by the primary coil and A_s is the cross-sectional area of the material in the core.

This secondary coil then allows us to calculate B by integrating $\mathcal{E} = -N_s \frac{d\Phi}{dt}$ to give:

$$B = -\frac{1}{N_s A_s} \int \mathcal{E} dt \quad (2.8)$$

Then, through the use of an integrator circuit, $V_{out} = -\frac{1}{R_i C} \int \mathcal{E} dt$ (where R_i is the integrator's input resistor, C the feedback capacitor and V_{out} the integrator output voltage), B is calculated using:

$$B = -\frac{R_i C}{N_s A_s} V_{out} \quad (2.9)$$

Our loop allows us to calculate the power loss per unit volume per cycle from the area under the $B - H$ curve.

$$\frac{P_{loss}}{V} = f \int H dB, \quad (2.10)$$

Where f is the frequency of the applied magnetic field.

This power loss is mainly due to two effects: hysteresis loss and eddy current loss. Hysteresis loss is the energy dissipated due to the lagging of the magnetic domains behind the applied field, while eddy current loss is due to circulating currents induced in the material by the changing magnetic field.

Further, μ_r values at different points on the curve can be calculated as linear dependence is assumed in the localised region.

$$\mu_r = \frac{1}{\mu_0} \frac{dB}{dH} \quad (2.11)$$

3 Experimental Methods

3.1 Coil assembly and primary circuit

Two coils, one of ~ 500 turns and one of ~ 400 turns, were co-axially wound into a solenoid (Length $L = 4.2 \pm 0.1$ cm). The coil with 500 turns was designated as the primary coil and connected in series with a $2 \pm 0.1 \Omega$ resistor (R_p) to reduce the current. From this, the current through the coil and thus the magnetic field strength through the coil ($I = \frac{V}{R_p} \rightarrow H = \frac{nI}{L}$, see Equation (2.1)) were calculated. This part of the circuit was completed by connecting it to a signal generator set to produce an AC voltage of $5 \pm 0.01 V_{pp}$ (Peak to Peak) at 50 ± 0.1 Hz (to emulate mains electricity supply).

The secondary coil was then attached to an integrator circuit.

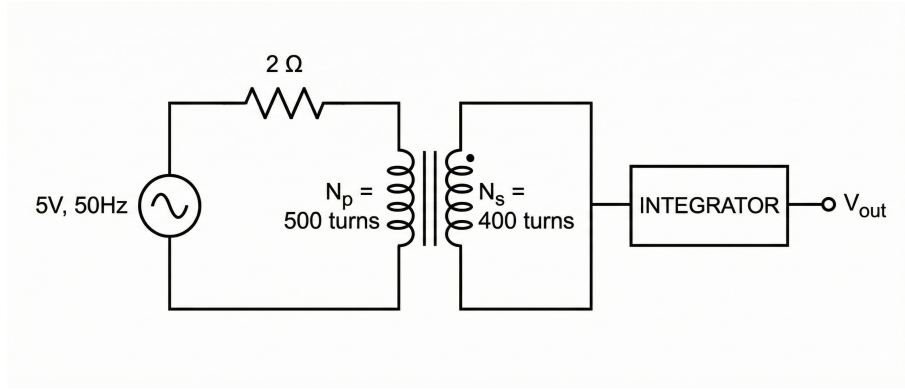


Figure 4: Integrator circuit.

3.2 Integrator design and calibration

The integrator circuit was set up as shown below:

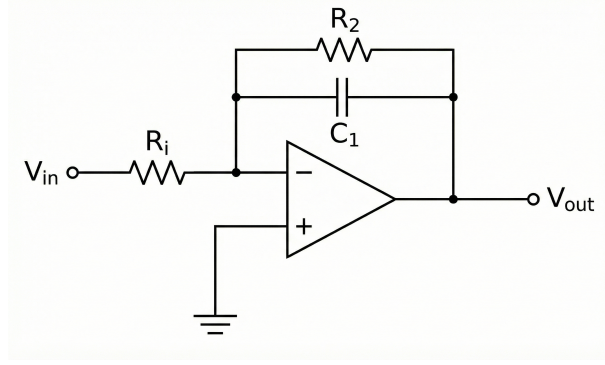


Figure 5: Integrator circuit.

R_2 was chosen to be $1 \text{ M}\Omega$.

After measuring R_2 , it was found to be $991.2 \pm 0.1 \text{ k}\Omega$.

The value of C_1 is chosen such that $Z_i \ll 1 \text{ M}\Omega$ at 50 Hz .

$$Z_i \ll 1 \text{ M}\Omega \quad (3.1)$$

$$Z_i = \frac{1}{\omega C_1} \quad (3.2)$$

$$\frac{1}{\omega C_1} \ll 1 \text{ M}\Omega \quad (3.3)$$

$$\frac{1}{2\pi \times 50 \times C_1} \ll 10^6 \quad (3.4)$$

$$C_1 \gg 3.18 \text{ nF} \quad (3.5)$$

Therefore, picking 100 nF was suitable.

To make the calculations easier, an R_i value that would give a gain of 1 was selected.

$$\text{gain} = \left| \frac{V_{out}}{V_{in}} \right| = \frac{1}{\omega R_i C_1} \quad (3.6)$$

$$1 = \frac{1}{\omega R_i C_1} \quad (3.7)$$

$$R_i = 31.83 \text{ k}\Omega \quad (3.8)$$

$$R_i \approx 30 \text{ k}\Omega \quad (3.9)$$

The chosen resistor and capacitor were then measured accurately, giving:

$$R_i = 32,830 \pm 5 \text{ }\Omega \quad (3.10)$$

$$C_1 = 97.8 \pm 0.1 \text{ nF} \quad (3.11)$$

$\therefore$

$$\text{theoretical gain} = \frac{1}{2\pi \times 50 \times 32,830 \times 97.8 \times 10^{-9}} = 0.9913 \quad (3.12)$$

After setting up the circuit, the oscilloscope was attached to the signal generator to measure the input and the output of the circuit.

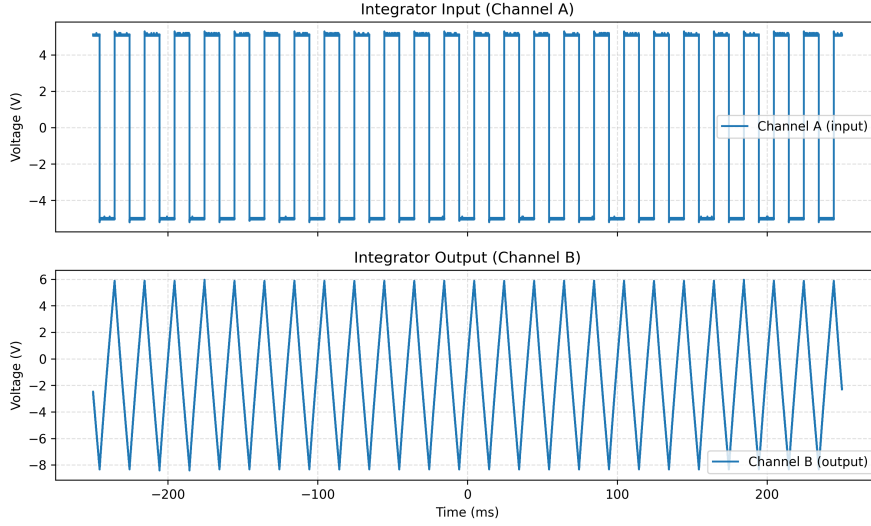


Figure 6: Measurements of the input and output of the circuit.

$$\text{Measured gain} = \left| \frac{V_{out}}{V_{in}} \right| = \frac{14.38V}{10.49V} = 1.37 \pm 0.005 \quad (3.13)$$

This corresponds to a 38% deviation from the theoretical value and may be due to the square wave that was tested. The square wave is composed of sinusoidal waves of many frequencies and therefore contains odd harmonics of the fundamental frequency of the integrator circuit (e.g. $3f = 150$ Hz, $5f = 250$ Hz, etc.). Combined with the op-amp not being ideal and having a limited bandwidth, these harmonics can result in a different gain than the theoretical value. Whilst this was due to systematic errors in the experiment, measured gain was used as gain in the subsequent calculations.

3.3 Data acquisition and sample preparation

The samples used were:

- Mild steel
- Transformer iron
- CuNi alloy at 10°C and above 40°C

The coil length and the cross-sectional area of the samples were measured. This was done using a vernier caliper to measure the length and width of the cuboid CuNi alloy and transformer iron samples and the diameter of the cylindrical mild steel sample.

Then the experiment was run, The signal generator was turned on to feed an AC current into the primary coil. After waiting until the circuit reached a steady state, the data points for 20 cycles were saved. This was done immediately to ensure the coil didn't heat up and change its resistance during the measurement. The coil was then turned off, and this was repeated for the other samples.

First, the measurements were performed without a sample (air) to act as a control, then metals.

When measuring the CuNi alloy, it was placed into hot water to raise its temperature to above 40°C and then placed into ice water to cool it to 10°C. The low time difference between the samples being removed from the water and being tested results in minor temperature changes during the measurement, but the actual temperature was not measured during the hysteresis and thus the temperatures should be taken as approximate.

Voltages across R_p (Channel A) and integrator output (Channel B) were recorded using a PicoScope at a sampling rate of 1.95 MHz (in spite of the AC current being used the PicoScope was set to DC. This was to avoid the possibility of some AC current being filtered out by the PicoScope as it is interpreted as DC offset due to the low frequency being used). For each sample, 20 cycles at 50 Hz (~ 0.32 s) were recorded.

3.4 Data processing

The recorded voltage data was processed using a Python script. This script performed the following tasks:

- split the data into 20 loops
- calculated H and B values for each loop using Equations (2.1) and (2.9)
- plotted the hysteresis loops
- performed a polynomial fit to resolve the loops at the ends of the H range
- calculated μ_r values using Equation (2.11)
- calculated power loss per unit volume using Equation (2.10)
- calculated mean and standard deviation values for $\mu_{r,max}$ and power loss over the 20 loops

4 Results and Analysis

4.1 Overview

The first step was to record the nominal measurements of the samples used throughout the run, which are summarised in Table 1 below. Over the course of the experiment five sets of data were collected:

- The air control experiment
- Mild steel
- Transformer iron
- CuNi alloy at approximately 10°C
- CuNi alloy above 40°C

The transformer iron and elevated-temperature CuNi alloy datasets were each recorded twice; the repeats agreed with the original measurements within the random scatter, so only the original runs are used in the tables below.

Table 1: Nominal sample dimensions and cross-sectional areas used to convert the voltage measurements into magnetic quantities.

Sample	Key measurements	Cross-sectional area (mm ²)
Air control	Diameter = 7.8 ± 0.1 mm	47.8 ± 1.2
Mild steel	Diameter = 3.12 ± 0.04 mm	7.65 ± 0.2
Transformer iron	width = 4.22 ± 0.02 mm, Thickness = 0.7 ± 0.02 mm	2.95 ± 0.16
CuNi alloy	Diameter = 5.00 ± 0.02 mm	19.6 ± 0.16

Twenty loops of the steady-state response were recorded for each dataset. Using Equations (2.1) and (2.9), the voltages across R_p and the integrator output were converted into magnetic field strength and magnetic flux density for the plots that follow.

From the multiple recorded loops, the mean and standard deviation of key metrics were calculated and used as estimates of random uncertainties. In all cases, loop-to-loop scatter was small. Larger systematic uncertainties are expected from calibration of the integrator gain, resistor values and sample geometry; these are not included in the quoted error bars but are discussed qualitatively in Section 5.

4.2 Air calibration

The air control plot shows the expected linear response with no hysteresis, since the permeability should be close to that of free space. The behaviour was modelled by $B = \mu_0 \mu_r H$, and the apparent relative permeability calculated from the slope is smaller than the expected value. The error in the fitted line does not account for this offset, so the discrepancy is interpreted as a systematic error possibly due to parasitic capacitance.

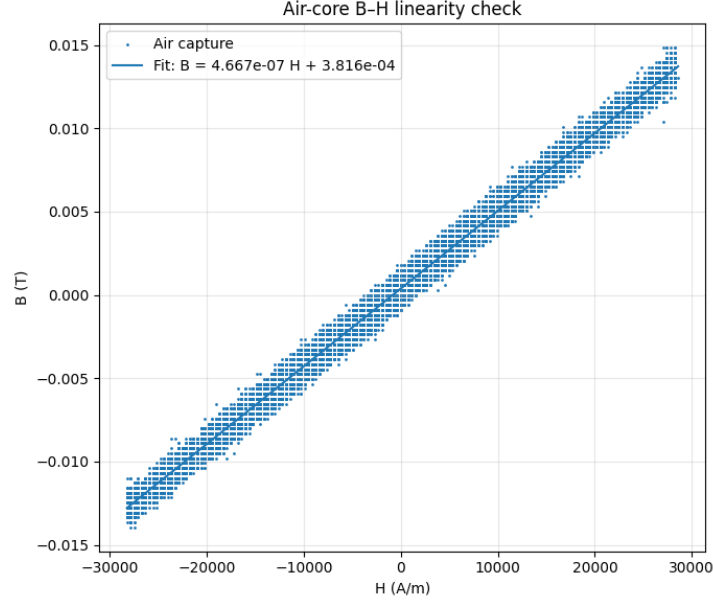


Figure 7: Linear $B - H$ response obtained from the air control experiment.

Figure 7 shows a strong linear response from the air control experiment as expected. There is minor scatter between the loops, which is likely due to noise in the measurement system. This resulted in the low uncertainty calculated using the standard deviation between gradients and intercepts in the set of modeled loops.

Interpreting the results from Figure 7, the relative permeability was calculated from equation 2.11.

$$\text{Ideally: } y = \frac{dB}{dH}x \quad (4.1)$$

$$\text{In reality: } y = (4.667 \times 10^{-07} \pm 1.260 \times 10^{-10})x + (3.816 \times 10^{-04} \pm 2.511 \times 10^{-06}) \quad (4.2)$$

Therefore:

$$\mu_r = 0.37 \pm 0.03 \quad (4.3)$$

This value is significantly lower than the expected $\mu_r \approx 1$ for air, indicating a systematic error in the measurement setup.

4.3 Hysteresis plots

The 20 recorded hysteresis loops for each material were plotted individually. In all three ferromagnetic datasets, the raw $B - H$ loops show small secondary loops at large $|H|$ (most clearly in transformer iron, Figure 10).

To account for this, the end tails of the hysteresis loops were fitted to fifth order polynomials allowing a smooth gradient to be found at the ends, allowing us to give a more stable estimate of the saturation permeability.

4.3.1 Mild steel

Figure 8 shows the raw hysteresis data for the mild steel sample.

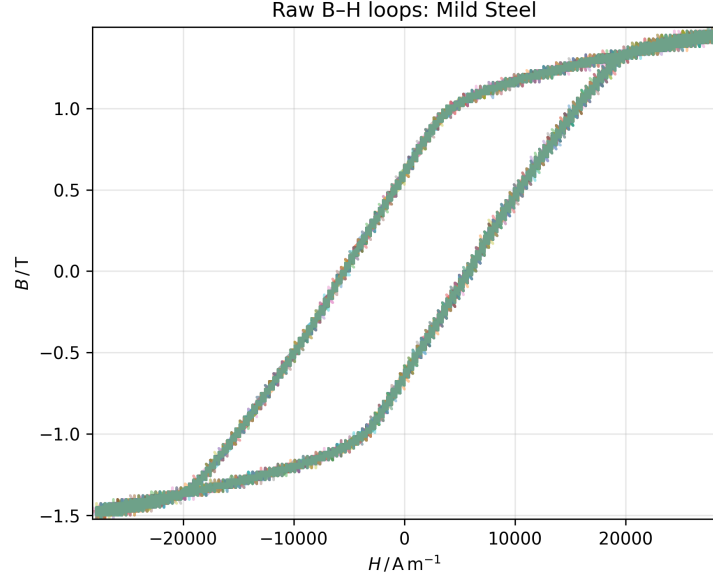


Figure 8: Raw mild steel hysteresis loops.

After applying the fifth-order polynomial fit to the tails, we obtained the smoother loop shown in Figure 9.

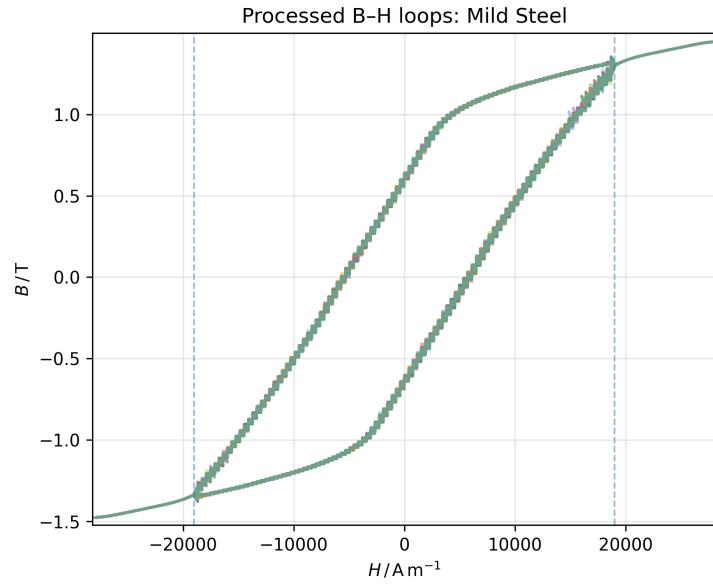


Figure 9: Mild steel hysteresis loop after polynomial tail fitting.

The processed loop (Figure 9) shows a relatively wide hysteresis curve with a saturation flux density of around 1.4T and a coercive field in the order of 10^3 Am^{-1} .

Table 2: Mild steel loop statistics from the polynomial-tail analysis.

Quantity	Value	Uncertainty	Units
B range	2.92	0.00034	T
H range	5.61×10^4	31	Am^{-1}
Power loss per unit volume	1.38×10^6	410	W m^{-3}
Maximum μ_r (numeric gradient)	123	4.5	—
Minimum μ_r (numeric gradient)	14.0	0.025	—

4.3.2 Transformer iron

Figure 10 shows the raw transformer iron loops, which also required polynomial smoothing of the tails.

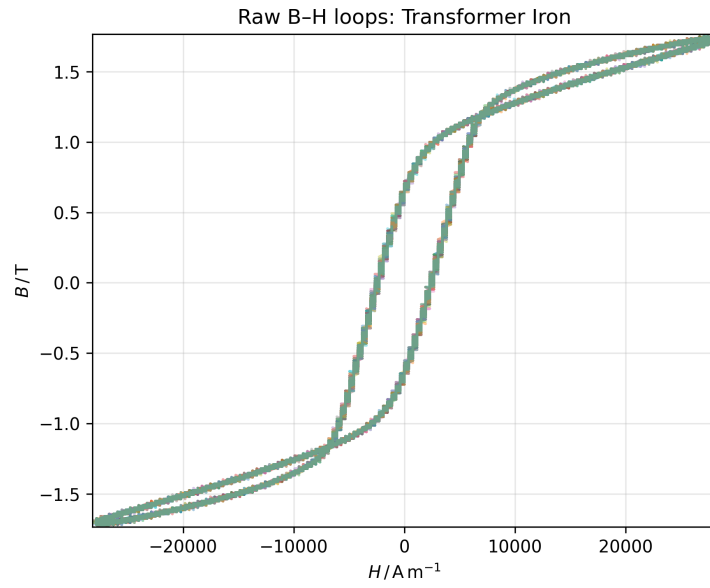


Figure 10: Raw transformer iron hysteresis loops.

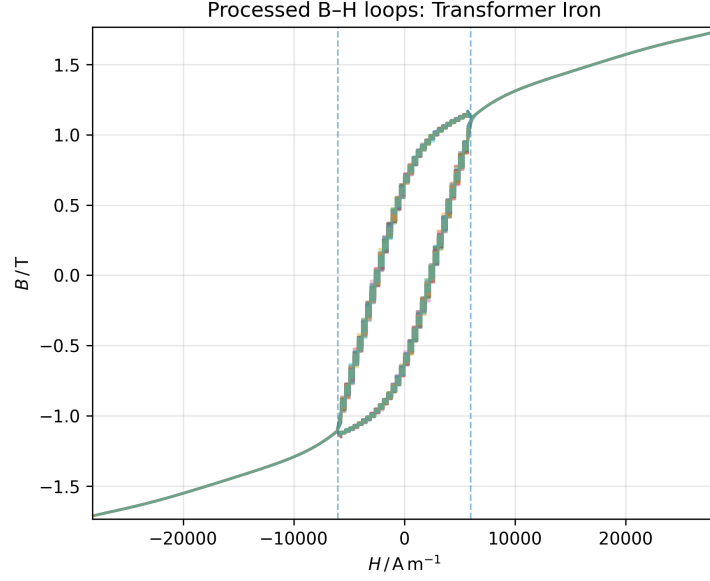


Figure 11: Transformer iron loop after polynomial tail fitting.

The transformer iron loop (Figure 11) is narrower and steeper in the central region than for mild steel, with a similar saturation flux density but a smaller coercive field.

Table 3: Transformer iron loop statistics from the polynomial-tail analysis.

Quantity	Value	Uncertainty	Units
B range	3.44	0.00068	T
H range	5.61×10^4	38	Am^{-1}
Power loss per unit volume	4.73×10^5	1050	W m^{-3}
Maximum μ_r (numeric gradient)	332	4.2	—
Minimum μ_r (numeric gradient)	13.9	0.025	—

4.3.3 CuNi at 10 °C

The CuNi alloy at 10 °C follows similar behaviour, with the raw data plotted in Figure 12.

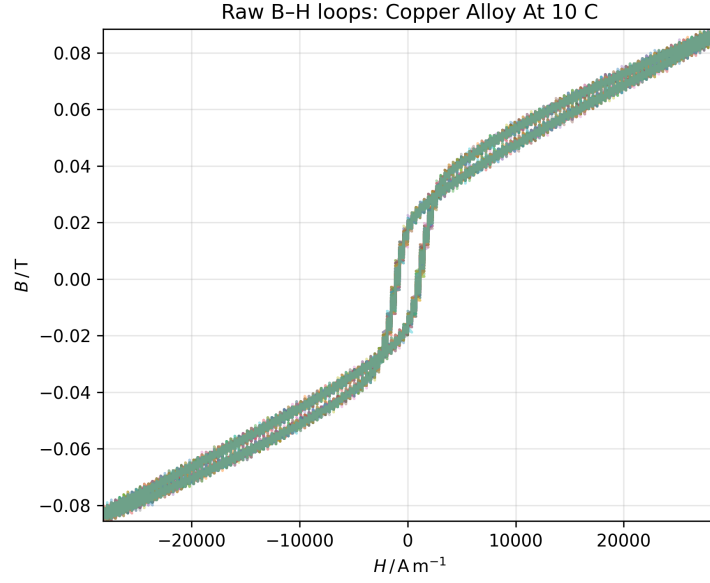


Figure 12: Raw CuNi (10 °C) hysteresis loops.

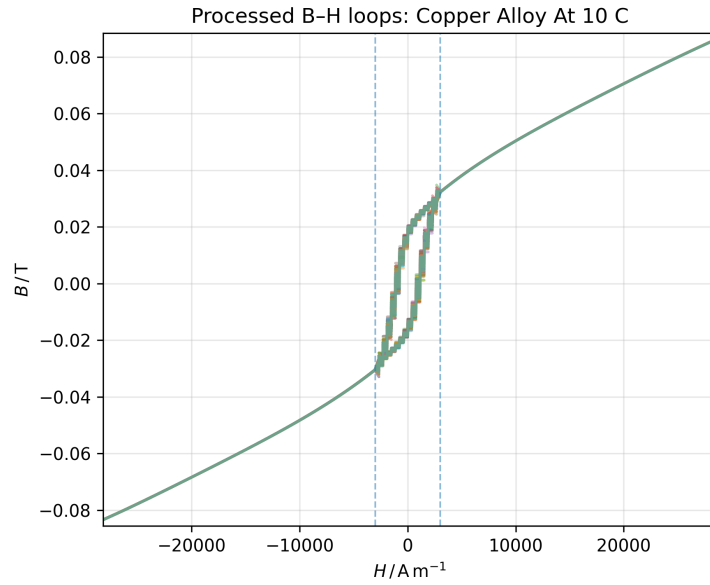


Figure 13: Polynomial-fit CuNi (10 °C) loop.

At 10 °C the CuNi alloy still shows a small but clear hysteresis loop (Figure 13), but the B range of only 0.17 ± 0.01 T is about 20 times smaller than for transformer iron, and the loop is much narrower in H .

Table 4: CuNi (10 °C) loop statistics after polynomial tail smoothing.

Quantity	Value	Uncertainty	Units
B range	0.17	0.00001	T
H range	5.68×10^4	1.7×10^{-12}	Am^{-1}
Power loss per unit volume	4605	20.5	W m^{-3}
Maximum μ_r (numeric gradient)	7.88	0.0023	—
Minimum μ_r (numeric gradient)	1.35	0.0037	—

4.3.4 CuNi above 40 °C

Heating the CuNi sample above 40 °C suppresses the hysteresis, leaving a near-linear $B - H$ response (Figure 14). A linear regression therefore replaced the polynomial tail fits for this dataset.

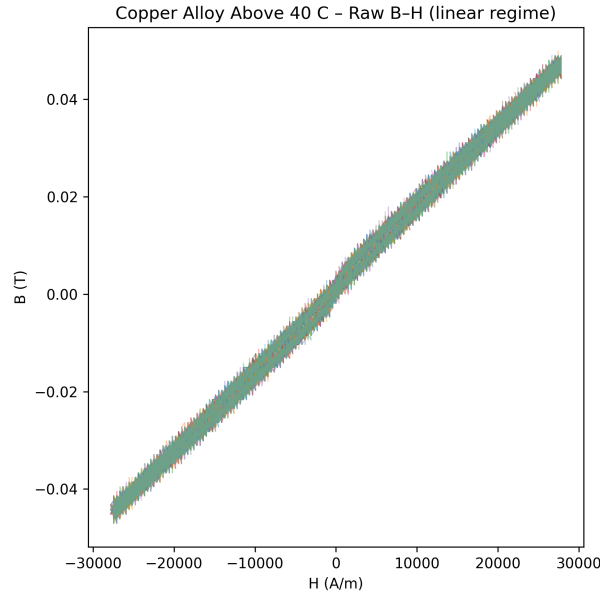


Figure 14: Raw CuNi (above 40 °C) data showing the predominantly linear response.

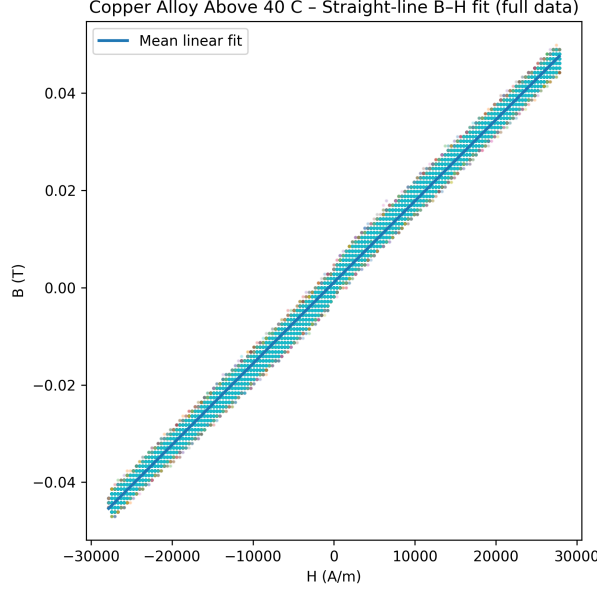


Figure 15: Linear regression for the CuNi (above 40 °C) run.

When the CuNi alloy is heated above 40 °C, the loop collapses into an almost straight line (Figure 15). Within the scatter, the data are consistent with a single linear relation $B = B_0 + (1.67 \pm 0.03) \times 10^{-6}H$, corresponding to an effective $\mu_r = 1.33 \pm 0.01$.

This is consistent with the alloy being just above its Curie temperature, where the material is only weakly paramagnetic and shows negligible hysteresis.

Table 5: Linear regression summary for CuNi above 40 °C.

Quantity	Value	Uncertainty	Units
B range	0.0954	0.00011	T
H range	5.56×10^4	19	Am^{-1}
Slope dB/dH	1.67×10^{-6}	3.36×10^{-10}	$\text{TA}^{-1}\text{m}^{-1}$
Intercept B_0	0.00108	6.58×10^{-6}	T
Relative μ_r (linear fit)	1.33	7.2×10^{-5}	—

4.4 Summary of results

Table 6: Summary of the collected data sets.

Data set	Notes
Air control experiment	Baseline linear response, no hysteresis expected.
Mild steel	Hysteresis loop with pronounced end tails.
Transformer iron	Initial loop deviated from literature; repeated run matched original.
CuNi alloy at 10 °C	Hysteresis loop with tail artefacts.
CuNi alloy above 40 °C	Repeat confirmed the linear response with negligible hysteresis.

Table 7 collects each sample’s key $B - H$ metrics, peak and minimum μ_r , and power loss per unit volume so the datasets can be compared directly.

Table 7: Aggregated magnetic metrics across the measured trials.

Sample	B range (T)	H range (A m ⁻¹)	Max μ_r	Min μ_r	Power loss per unit volume (W m ⁻³)
Mild steel	2.92	5.61×10^4	123	14.0	1.38×10^6
Transformer iron	3.44	5.61×10^4	332	13.9	4.73×10^5
CuNi (10 °C)	0.17	5.68×10^4	7.88	1.35	4605
CuNi (above 40 °C)	0.0954	5.56×10^4	1.33	1.33	negligible

Comparing the samples, transformer iron has the largest maximum permeability ($\mu_{r,\max} \approx 332$) and the smallest hysteresis loss ($\sim 5 \times 10^5 \text{ W m}^{-3}$), making it the most suitable transformer-core material among those tested. Mild steel also has a relatively high μ_r but suffers from much larger hysteresis losses. The CuNi alloy has much lower μ_r and loss; at 10 °C it shows a small hysteresis loop, while above 40 °C it behaves almost linearly with $\mu_r \approx 1.3$, i.e. only slightly different from air.

5 Discussion

5.1 Recap of key trends

The experiment aimed to use the B – H curves of different materials to quantify which would be the best for use as the core of a transformer iron. This was done by comparing their maximum relative permeability $\mu_{r,\max}$, power loss per cycle, and other associated factors calculated from their B – H curves.

The relative permeabilities indicated the ranking:

transformer iron > mild steel $\gg$ CuNi

While the energy loss per cycle followed:

mild steel > transformer iron $\gg$ CuNi.

Both trends are coherent with the behaviour expected from the literature.

Table 8: Summary of key magnetic metrics extracted from the B – H loops.

Material	$\mu_{r,\max}$	Power loss per cycle (W m ⁻³)	Notes
Transformer iron	332	4.73×10^5	Narrow, steep loop
Mild steel	123	1.38×10^6	Wider hysteresis
CuNi (10 °C)	7.9	4.6×10^3	Very low loss but low μ_r
CuNi (above 40 °C)	1.33	negligible	Loop collapses

5.2 Air control results

As described in Section 4.2, for air we expect $\mu_r \approx 1$ and a straight line through the origin (Eq. (2.4)). The air-control fit shows a residual intercept (Eq. (4.2)) and yields a measured $\mu_r = 0.37 \pm 0.03$ (Eq. (4.3)). This shows the apparatus is systematically mis-calibrated.

5.3 Which core is best and why?

An ideal core has a large μ_r in its working region and a low energy loss per cycle. A high μ_r allows the core to efficiently produce a high magnetic flux density, reducing magnetising current for a given H . Transformer iron is therefore the ideal choice: it has $\mu_{r,\max} \approx 332$, roughly three times that of mild steel (≈ 123), and a much smaller power loss per unit volume ($4.73 \times 10^5 \text{ W m}^{-3}$).

The narrower and steeper loop expected for transformer cores is illustrated schematically in Figure 11, reflecting rapid domain alignment and a large working range. Within the accuracy of this experiment, transformer iron is clearly the most suitable of the ferromagnetic materials. Mild steel would incur significantly larger hysteresis losses under similar magnetising conditions.

Now, comparing it to the CuNi alloy at 10 °C, CuNi has $\mu_{r,\max} \approx 8$ and very low losses, but the low permeability would require much higher H to achieve the same B , making it a poor transformer core material.

5.4 Minimum μ_r values for transformer cores

The minimum μ_r values seen were:

- Transformer iron: $\mu_{r,\max} \approx 13.9$
- Mild steel: $\mu_{r,\max} \approx 14.0$
- CuNi (10 °C): $\mu_{r,\max} \approx 1.35$

The μ_r values are expected to approach 1 as H increases, since the domains become fully aligned. The large μ_r values seen show that the AC current switches before saturation is reached.

5.5 Temperature dependence and Curie-like behaviour of CuNi

Beyond the low μ_r value, there are issues with the temperature range of the CuNi alloy. Prolonged transformer use would heat the core, exceeding the above-40 °C condition tested in Figure 15.

A shift from 10 °C to above 40 °C is shown to dramatically change the magnetic properties:

- 10 °C: small but clear hysteresis loop; B -range ≈ 0.17 T; $\mu_{r,\max} \approx 7.9$; power loss $\approx 4.6 \times 10^3$ W m⁻³.
- Above 40 °C: loop collapses to a straight line; almost no hysteresis; $\mu_r \approx 1.33$ nearly constant over the measured range.

This behaviour reflects the alloy approaching or exceeding its Curie temperature. This is the point at which the thermal vibration of atoms disrupts the alignment of magnetic domains, so the material becomes effectively paramagnetic.

Table 9: Temperature dependence of CuNi hysteresis metrics.

Temperature (°C)	B -range (T)	$\mu_{r,\max}$	Power loss (W m ⁻³)
10	0.17	7.9	4.6×10^3
above 40	0.0954	1.33	negligible

5.6 Uncertainties, limitations, and their impact

Random errors were present during the experiment:

Loop-to-loop scatter was small; the standard deviations in Tables 2 – 5 are tiny compared with the mean values. This means random uncertainties in $\mu_{r,\max}$ and loop area are negligible.

In addition, there were a number of key **systematic errors**:

- **Integrator gain:** measured gain (≈ 1.37) versus theoretical (≈ 0.99) differed by $\sim 40\%$, propagating directly into all B and μ_r values due to resistance and winding-count uncertainties.
- **Geometry / area assumptions:** the sample did not fill the coil, so assuming flux is dominated by A_{sample} underestimates B_{sample} and biases μ_r low.
- **Finite frequency and parasitic effects:** the integrator was designed for 50 Hz, but frequency-dependent response, square-wave calibration harmonics, parasitic capacitance, and coil resistance distort waveforms.
- **Temperature measurement for CuNi:** sample temperatures during hysteresis runs were not directly measured; only approximate starting values ($\approx 10^\circ\text{C}$ and $> 40^\circ\text{C}$) were noted, and the sample likely drifted between bath and measurement.

All these effects result in the raw measurements being off by a multiplicative amount. This has led to unreliable calculated values for the properties of the materials. However, all the materials were subject to the same systematic errors and thus the relative measurements and the resulting ranking of the materials should hold up to scrutiny.

5.7 Possible improvements and future work

Future experiments could reduce systematic uncertainties and map the material behaviour more completely by:

- Calibrating the integrator with a pure sinusoid at 50 Hz and known amplitude rather than a square wave with harmonics.
- Using samples that fill the entire coil cross-section to minimise flux fringing and geometry errors.
- Measuring sample temperature directly (e.g., a thermocouple attached to the rod) and acquiring hysteresis loops across a finer temperature grid to capture the CuNi transition.
- Recording additional datasets at different frequencies to study the scaling of eddy-current losses with f .

These improvements would decrease systematic uncertainty and clarify how each material's intrinsic properties affect transformer-core suitability.

6 Conclusion

This experiment compared the $B - H$ responses of several candidate transformer-core materials when driven by the same AC current, with the aim of identifying the sample that simultaneously maximises relative permeability and minimises hysteresis loss. Each dataset was processed into a hysteresis loop, which then yielded maximum μ_r and cycle loss estimates (Table 7). Transformer iron clearly outperforms the other ferromagnets, achieving $\mu_{r,\text{max}} \approx 3.3 \times 10^2$ and only $\sim 4.7 \times 10^5 \text{ W m}^{-3}$ of loss, while the CuNi alloy remains close to air ($\mu_r \approx 1.3$) once the temperature is raised above 40°C , consistent with a transition through its Curie point.

The ideal transformer core therefore has a high local μ_r to deliver large flux densities for modest H fields and a narrow loop to keep the cycle loss low. Within the constraints of the apparatus, transformer iron best satisfies both criteria, whereas mild steel sacrifices efficiency through its wider hysteresis and CuNi lacks magnetic hardness or permeability for practical transformer operation.

Several systematic issues limit the absolute accuracy of the reported values: the integrator gain differed from its design by about 40 %, the samples did not completely fill the measurement coil, and parasitic elements distort the waveform harmonics. These effects bias every dataset by a similar multiplicative factor, so the relative rankings reported here still reflect the intrinsic behaviour of the materials even if the magnitudes carry larger uncertainty.

Future work should therefore focus on refining the calibration loop (pure 50 Hz sinusoid with known amplitude), using samples that match the coil geometry, and extending the temperature/frequency matrix for CuNi to pin down its Curie-like transition with greater fidelity.

References

- [1] Robert Keim. What is hysteresis? an introduction for electrical engineers, 2023.
- [2] geeksforgeeks. Hysteresis loop, 2025.